Reverse of differentiation

1

a $4x^3$

b $x^4 + C$

2

a $6x^5$

b $4x^6 + C$

3

a $5x^4 + 4x^3$

b $7x^5 + 7x^4 + C$

4

a $-4x^{-5}$

b $-4x^{-4} + C$

5

a $8x^7 + 6x^5$

b $\frac{1}{2}x^8 + \frac{1}{2}x^6 + C$

6

a $\frac{6}{5}\sqrt[5]{x}$

b $5\sqrt[5]{x^6} + C$

Anti-differentiation

7

a $f(x) = 2x + C$

b Linear

c $f(x) = mx + C$

8

a $f(x) = -4x^2 + C$

b Quadratic

c $f(x) = \frac{a}{2}x^2 + C$

9

a $f(x) = 2x^2 + 3x + C$

b Quadratic

c $f(x) = \frac{a}{2}x^2 + bx + C$

10

a $f(x) = \frac{7x^3}{3} + C$

b Cubic

c $f(x) = \frac{a}{3}x^3 + C$

11

a $y = 9x + C$

b $y = 4x^2 + C$

c $y = 5x^2 + 7x + C$

d $y = 3x^3 + C$

e $y = 3x^3 + 2x^2 - 6x + C$

f $y = \frac{x^6}{6} + C$

g $y = 3x^5 + 4x^4 + C$

h $y = \frac{x^7}{28} + \frac{x^3}{9} + C$

i $y = -\frac{x^{-5}}{5} + C$

j $y = -\frac{3}{x^5} + C$

k $y = -\frac{2}{x^5} + \frac{3}{x^3} + C$

l $y = \frac{20x^{\frac{7}{5}}}{7} + \frac{21x^{\frac{11}{7}}}{11} + C$

m $y = \frac{7x^{\frac{4}{7}}}{4} + \frac{5x^{\frac{3}{5}}}{3} + C$

n $y = 2x^4 + \frac{9x^{\frac{8}{3}}}{8} - 3x + C$

o $y = x - 2x^{-2} + C$

p $y = 2x^5 - \frac{9x^4}{4} + C$

q $y = 5x^3 - 13x^2 + 8x + C$

r $y = \frac{x^3}{3} + 5x^2 + 24x + C$

s $y = \frac{2x^{\frac{3}{2}}}{3} + C$

t $y = 12\sqrt{x^3} + C$

u $y = \frac{2x^{\frac{5}{2}}}{5} + C$

v $y = 12\sqrt{x} + C$

Applications

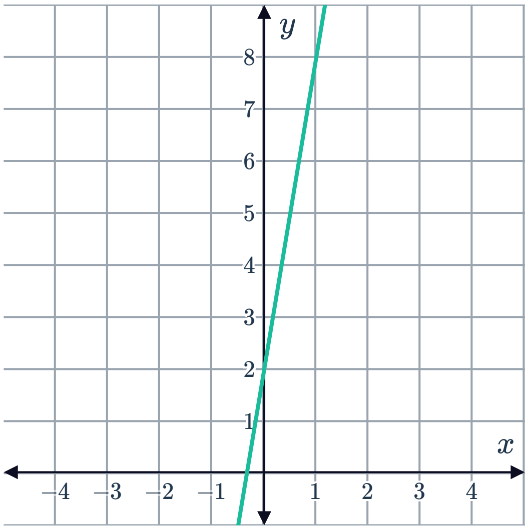
12

a

i Linear

ii $f(x) = 6x + C$

iii Any straight line with gradient of 6:

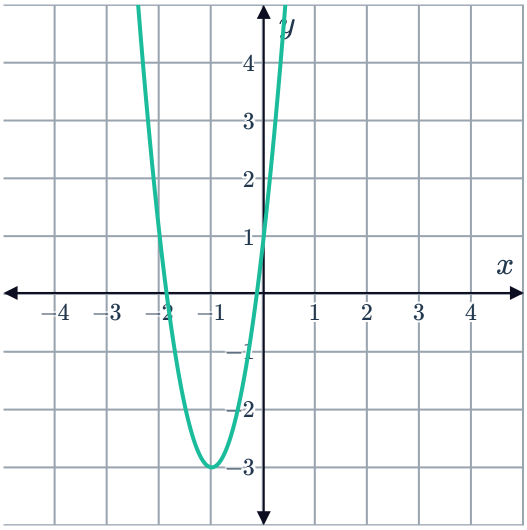


b

i Quadratic

ii $f\left(x\right) = 4x^2 + 8x + C$

iii Any parabola of the form $f'\left(x\right) = 4x^2 + 8x + C$ with axis of symmetry $x = -1$:

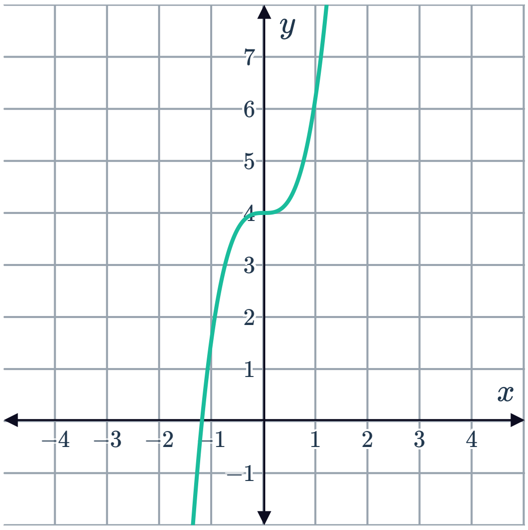


c

i Cubic

ii $f(x) = \dfrac{7x^3}{3} + C$

iii Any cubic of the form $\dfrac{7x^3}{3} + C$:

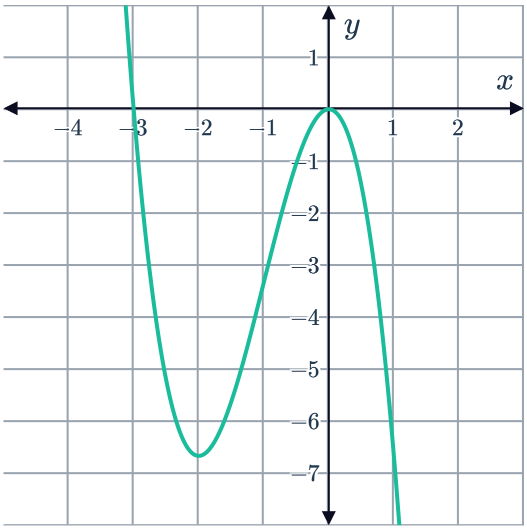


d

i Cubic

ii $f(x) = -\dfrac{5x^3}{3} - 5x^2 + C$

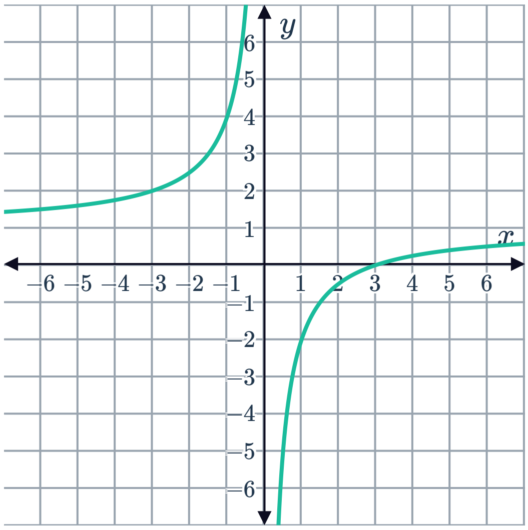
iii Any cubic of the form $f(x) = -\dfrac{5x^3}{3} - 5x^2 + C$:



13

a $f\left(x\right)=-\dfrac{3}{x}+C$

b Any graph of a rational function of the form $f\left(x\right)=-\dfrac{3}{x}+C$ with vertical asymptote at $x=0$ and horizontal asymptote at $y=C$:

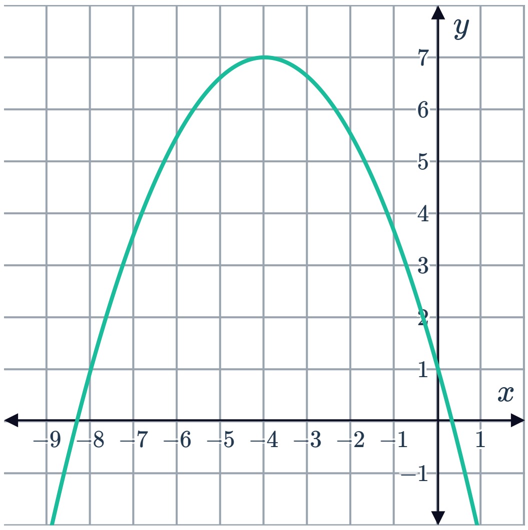


14

a $f'(x)=-\dfrac{3x}{4}-3$

b $f(x)=-\dfrac{3x^2}{8}-3x+C$

c Any parabola of the form $f(x)=-\dfrac{3x^2}{8}-3x+C$ with axis of symmetry $x=-4$:



Find value of C

15

a $y = 2x^2 + 7x + C$

b $y = 2x^2 + 7x + 2$

16

a $y = 3x^3 - 5x^2 + 2x + C$

b $y = 3x^3 - 5x^2 + 2x + 5$

17

a $y = 5x^3 + 7x + C$

b $y = 5x^3 + 7x + 5$

18

a $y = 2x^5 + 5x^4 + 2x^3 + 3x^2 + 9x + C$

b $y = 2x^5 + 5x^4 + 2x^3 + 3x^2 + 9x + 2$

19

a $y = \dfrac{27x^{\frac{5}{3}}}{5} + C$

b $y = \dfrac{27x^{\frac{5}{3}}}{5} + 5$

20 $p = 3t^2 - 5t + 3$

21 $y = 3x + 2$